ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 3: Pair of Linear Equations in Two Variables

Key Concepts & Important Formulas:

* Standard Form: a1*x + b1*y + c1 = 0 and a2*x + b2*y + c2 = 0.

* Consistency and Conditions:
- If a1/a2 != b1/b2: Intersecting lines, unique solution, consistent.
- If a1/a2 = b1/b2 = c1/c2: Coincident lines, infinitely many solutions, consistent (dependent).
- If a1/a2 = b1/b2 != c1/c2: Parallel lines, no solution, inconsistent.

* Algebraic Methods: 1. Substitution Method, 2. Elimination Method, 3. Cross-Multiplication Method.

Solved Exam Examples & Practice:

Example: Solve x + y = 14 and x - y = 4 by elimination.
Adding both equations: 2x = 18 => x = 9.
Substituting: 9 + y = 14 => y = 5.

Instructor: Mandeep Sir (Green Board) | Video lectures available on Root Classes | Price: FREE